# The Interaction Law: Capture Boundaries, Collapse Thresholds, and Scattering in Two-Core Event Density Dynamics
**(Interaction Laws III)**

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## Abstract
This paper presents the first complete interaction law for two-core encounters in the Event Density (ED) field. Using the standard shared-field Scenario H configuration, we systematically vary approach velocity and impact parameter to map the full MERGE/BIND landscape. The results reveal a clean architectural structure: a critical velocity below which merging is impossible; a monotone velocity-dependent capture boundary $b_c(v)$ separating MERGE from BIND; and a geometric collapse threshold at $D$min ≈ 23–24 px that governs identity loss independent of collision parameters. In the BIND regime, the surviving identities follow a smooth scattering law in which the deflection angle $\theta(b,v)$ is symmetric, peaks near the capture boundary, increases with velocity, and decays at large impact parameter. A left–right symmetry test confirms that these behaviors arise from the ED dynamics rather than numerical artifacts, with the only asymmetry attributable to a coordinate-dependent classifier label. Together, these findings establish the ED interaction law: a structured, symmetric, and geometrically grounded set of principles that organize all two-core encounters and provide the foundation for higher-order multi-core dynamics explored in subsequent work.

## 1. Introduction
Two-core interactions are the simplest nontrivial setting in which the Event Density (ED) field reveals its dynamical structure. Earlier papers in the ED-Arch series established the geometry of individuation, the stability of core identities, and the shared-field mechanisms through which cores influence one another. What remained unresolved was the nature of pairwise interaction: when two individuated cores approach each other in a shared field, what determines whether they merge, bind, scatter, or pass through? What governs the transition between these regimes? And is there an underlying architectural law that organizes these outcomes?

ED-Arch-12 addresses these questions directly. Using the standard two-core shared-field experiment, we systematically vary the approach velocity and impact parameter to map the full interaction landscape. The results reveal a clean and surprisingly structured law: a critical velocity below which merging is impossible, a monotone capture boundary separating MERGE from BIND, a geometric collapse threshold that governs identity loss, and a scattering law that describes deflection in the BIND regime. A symmetry test confirms that these behaviors arise from the ED dynamics themselves rather than numerical artifacts.

This paper presents the first complete interaction law in ED, establishing the architectural principles that govern how individuated cores influence, deflect, and sometimes collapse into one another.

## 2. Experimental Setup

### 2.1 Shared-field two-core scenario
All experiments use the standard Scenario H configuration: two individuated ED cores placed symmetrically on opposite sides of the domain, embedded in a shared field, and advected toward each other with equal and opposite velocities. The impact parameter $b$ controls the lateral offset between their approach trajectories. The shared field

mediates the interaction, producing curvature-well forces that bend trajectories, compress identities, or collapse them entirely.

The simulation runs for 400 steps or until the outcome is unambiguously determined. No code modifications are made at any stage; all results rely on the existing ED implementation, including its advection kernel, soft partition, blob detector, and classification logic.

### 2.2 Parameters

Two parameters define each run:

- Velocity $v$: the magnitude of the approach speed
- Impact parameter $b$: the lateral offset between the two cores' initial positions

For each pair $(v,b)$, the system evolves under the shared-field dynamics. The parameter sweeps span:

- coarse exploration of $(b,v)$
- fine zoom-in near the MERGE/BIND boundary
- bracketing of low- and high-velocity regimes
- a structured grid for the phase diagram
- a matching grid for the scattering map

### 2.3 Measurements

Three quantities are extracted from each run:

- Minimum separation $D_{\min}$: the closest approach between the two core centroids
- Deflection angles $\theta_A,\theta_B$: the angular deviation of each core's trajectory from its initial axis
- Outcome classification: MERGE, BIND, PASS, or SCATTER, determined by the existing classifier

MERGE is detected topologically through identity collapse in the blob detector. BIND corresponds to two surviving identities that deflect symmetrically. PASS indicates that the cores exchange lateral ordering without merging. SCATTER is rare in this parameter range and does not appear in the results.

## 3. Coarse Sweep: First Evidence of the Boundary

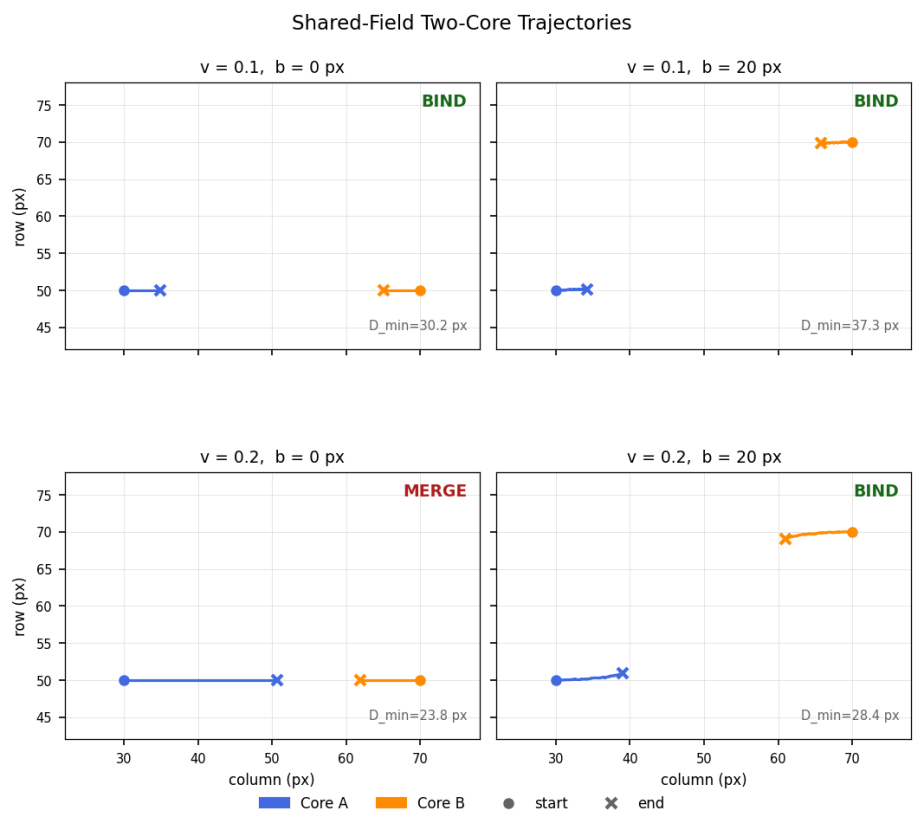
The first experiment probes the interaction landscape at a coarse resolution, using:

- Impact parameters: $b = 0,10,20$
- Velocities: $v = 0.1,0.2$

This sweep immediately reveals the qualitative structure of the interaction law.

At v = 0.1, all runs result in BIND: the cores approach, deflect symmetrically, and separate with their identities intact. At v = 0.2, the behavior changes sharply: head-on and small-offset collisions produce MERGE, while larger offsets still produce BIND. This is the first appearance of a MERGE/BIND transition, suggesting the existence of a critical impact parameter $b_c(v)$.

**Figure 1:** Coarse Interaction Outcomes Across (b, v)




Coarse sweep over (b = 0, 10, 20; v = 0.1, 0.2) showing the first appearance of the MERGE/BIND transition. At v = 0.1 all interactions result in BIND, while at v = 0.2 small-offset collisions produce MERGE and large-offset collisions produce BIND. This figure establishes the existence of a velocity-dependent capture boundary.

Two structural signatures emerge even at this early stage:

1. Geometric collapse threshold:

- All MERGE cases reach a minimum separation of approximately 23–24 px, independent of b.
- This suggests a geometric identity-collapse radius intrinsic to the ED field.

2. Symmetry vs asymmetry:

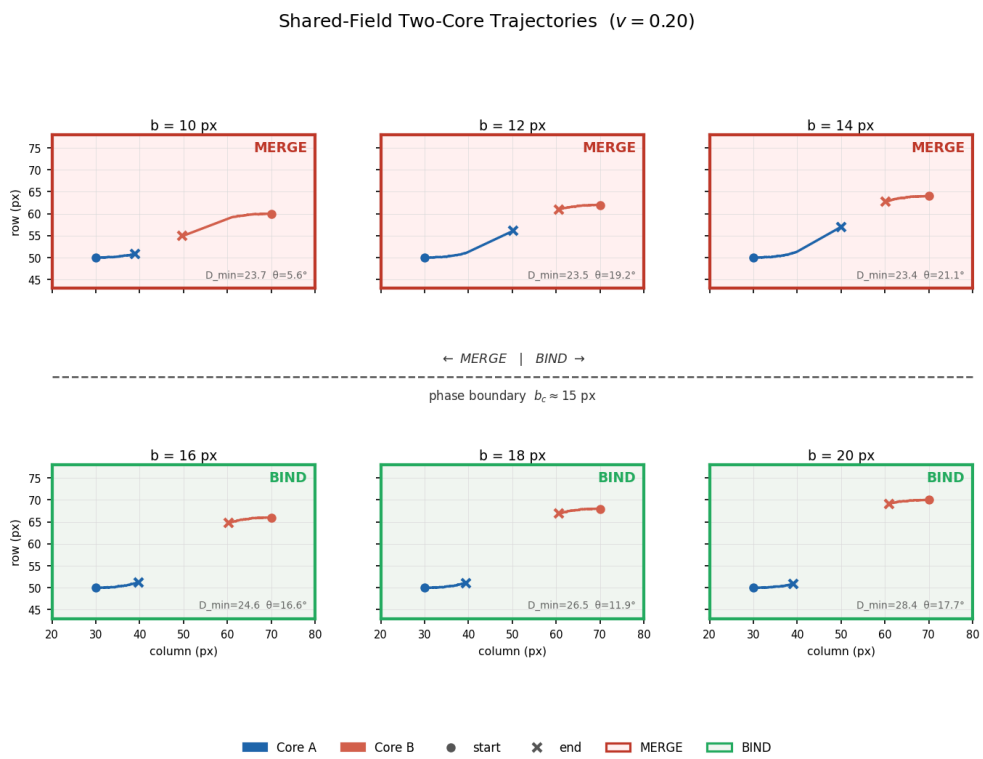
- BIND cases show symmetric deflection: $\theta_A = \theta_B$.
- MERGE cases show asymmetric deflection: one core's trajectory collapses to zero angle as its identity dissolves.

These observations motivate the finer sweeps that follow, ultimately revealing the full interaction law.

## 4. Boundary Zoom-In at v = 0.20

The coarse sweep revealed a MERGE/BIND transition near $b \approx 15$ at $v = 0.20$. To resolve this boundary precisely, we performed a fine-grained sweep over $b = 10,12,14,16,18,20$ at fixed velocity.

**Figure 2:** Fine Sweep at v = 0.20 Revealing $b_c \approx 15$ px




Fine-grained sweep at v = 0.20 showing MERGE for b = 10, 12, 14 and BIND for b = 16, 18, 20. The transition between b = 14 and b = 16 defines the critical impact parameter $b_c(0.20) \approx 15$ px. MERGE cases exhibit identity collapse, while BIND cases show symmetric deflection.

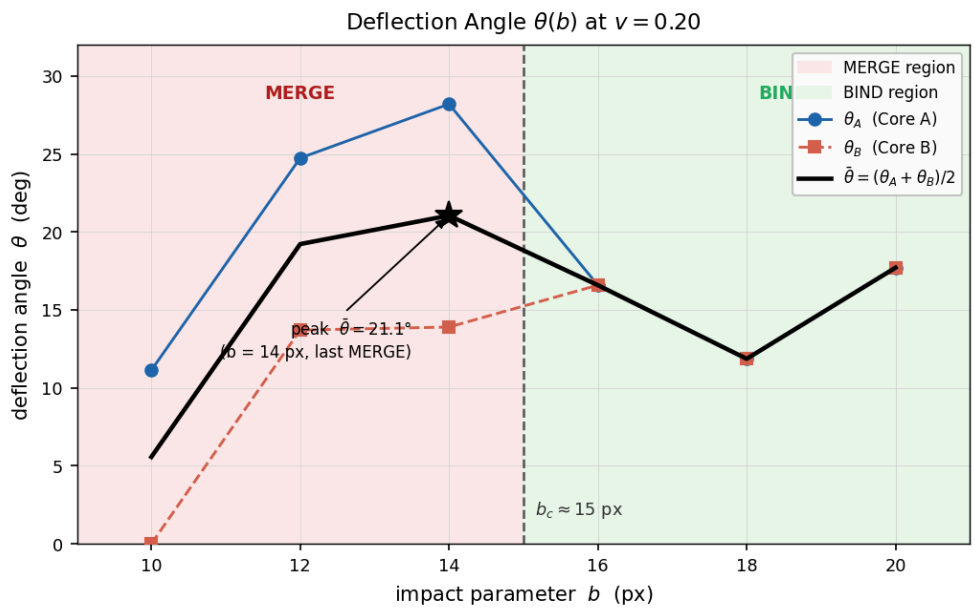
The results show a clean and sharp transition. MERGE occurs at $b$ = 10,12,14, while BIND occurs at $b$ = 16,18,20. The critical impact parameter is therefore:

$$b_c(0.20) \approx 15 \text{ px}.$$

Two structural features emerge clearly in this zoom-in:

(1) Peak deflection at the boundary.
The largest deflection angle occurs at $b$ = 14, the last MERGE case. This is the point where the curvature-well influence is strongest without yet triggering identity collapse. Immediately beyond the boundary, in the BIND regime, the deflection becomes symmetric and slightly smaller.

**Figure 3:** Peak Deflection at the MERGE/BIND Boundary




Deflection angle θ as a function of impact parameter at v = 0.20. The maximum deflection occurs at b = 14, the last MERGE case before identity collapse. This peak marks the point of maximum curvature-well influence without triggering MERGE.

(2) Geometric collapse threshold.
All MERGE cases reach a minimum separation of approximately 23–24 px. This value is nearly invariant across b and v, suggesting that identity collapse is triggered when the shared-field overlap exceeds a geometric threshold intrinsic to the ED field. The boundary sweep confirms that MERGE is not a gradual process but a topological transition that occurs when the system crosses this collapse radius.

This zoom-in establishes the existence and location of the MERGE/BIND boundary at $v = 0.20$, and it motivates the broader mapping of $b_c(v)$ across velocities.

## 5. Velocity Bracketing

To understand how the capture boundary depends on velocity, we probed two additional regimes: a low-velocity regime where merging might be impossible, and a high-velocity regime where merging might dominate.

### 5.1 Low-velocity regime (v = 0.15)

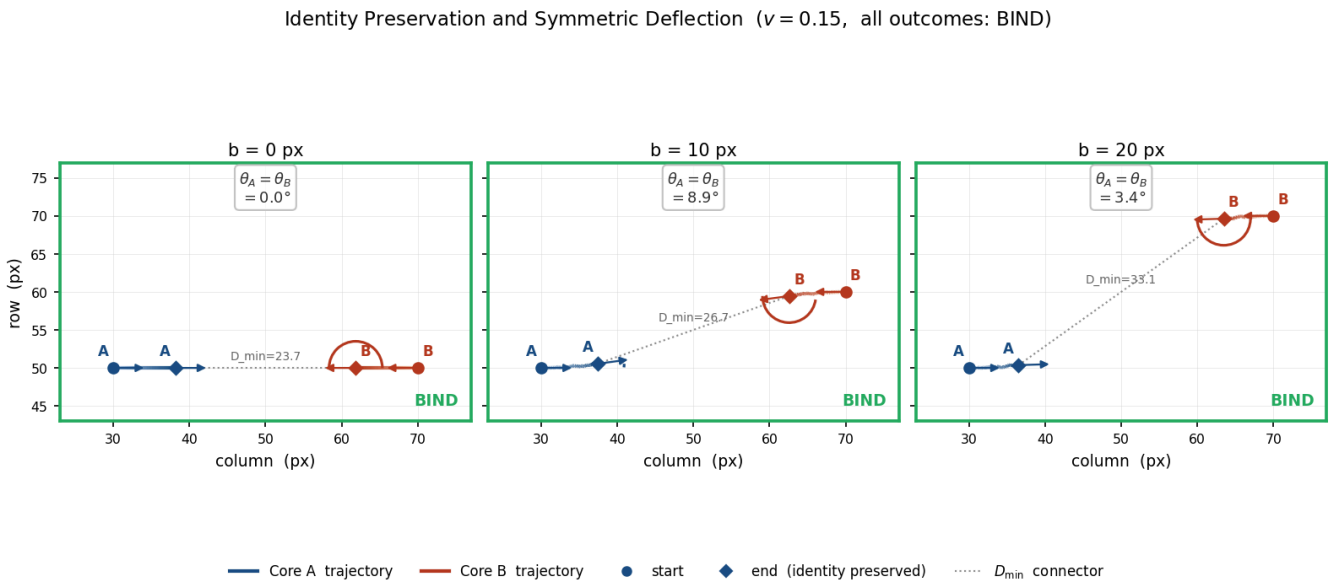
At $v = 0.15$, all runs— including the head-on case $b = 0$—result in BIND. The cores approach to the same geometric floor (D_min ≈ 23.7 px) seen in MERGE cases at higher velocities, but at this lower speed the system relaxes and the identities re-separate. No amount of impact-parameter tuning produces a merge.

This establishes that:

$$v = 0.15 \text{ is subcritical.}$$

The critical velocity for merging lies between $v = 0.15$ and $v = 0.18$.

**Figure 4:** Subcritical Velocity: v = 0.15 Produces Only BIND




Representative trajectories at v = 0.15 showing that all interactions result in BIND, including the head-on case. Although the cores approach the geometric collapse distance, they re-separate, demonstrating that v = 0.15 is below the critical velocity for merging.

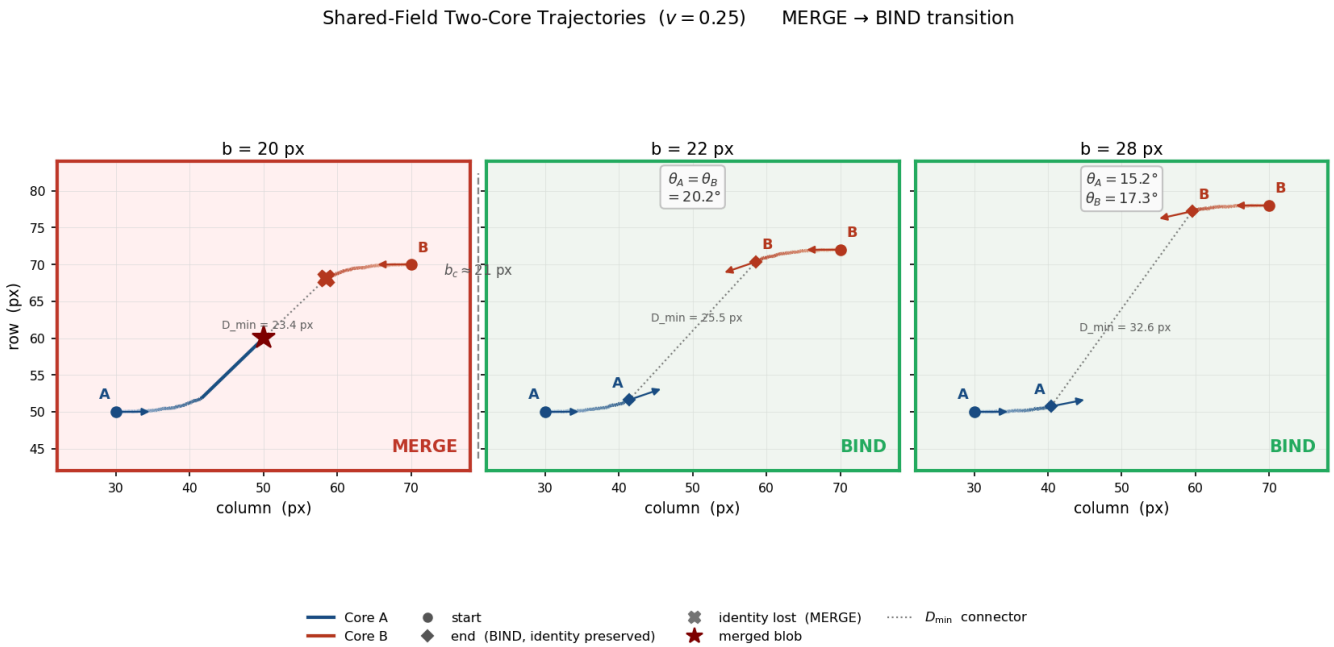
### 5.2 High-velocity regime (v = 0.25)

At $v = 0.25$, the opposite pattern emerges. MERGE occurs even at relatively large offsets (e.g., $b = 20$), and BIND only appears once the impact parameter is increased beyond $b = 22$. This yields:

$$b_c(0.25) \approx 21 \text{ px}.$$

The MERGE D_min values remain tightly clustered around 23–24 px, reinforcing the geometric collapse threshold observed earlier.

**Figure 5:** High-Velocity Regime: v = 0.25 Expands the MERGE Region




Representative trajectories at v = 0.25 showing MERGE at moderate offsets (e.g., b = 20) and BIND only at large offsets (b ≥ 22). The expansion of the MERGE region with velocity illustrates the monotone growth of the capture boundary $b_c(v)$.

### 5.3 Monotone capture boundary

Combining the low-, mid-, and high-velocity results reveals a monotone relationship:

- $b_c(0.15) < 0$ (no MERGE at any b)
- $b_c(0.18) \approx 12$–$13$
- $b_c(0.20) \approx 15$
- $b_c(0.22) \approx 17$–$18$
- $b_c(0.25) \approx 21$

The capture boundary grows steadily with velocity, expanding the MERGE region in (b, v) space. The approximate slope is about +5 px per +0.05 in v across this range.

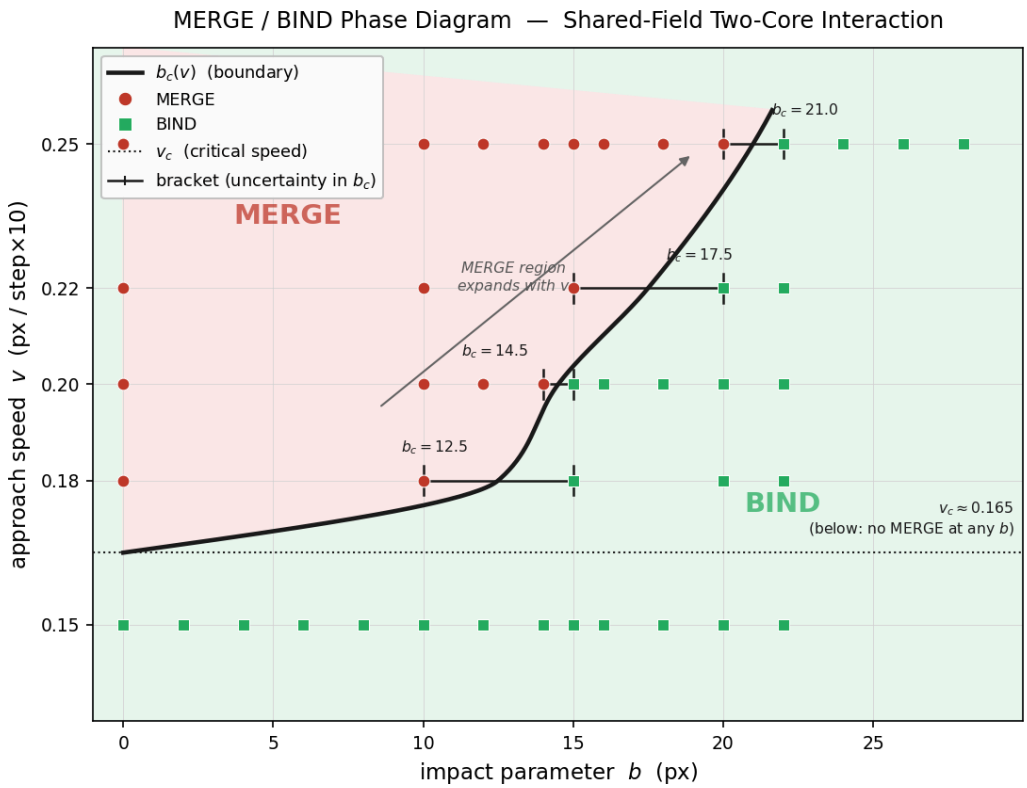
This monotonicity is a hallmark of an underlying architectural law: faster approaches allow the system to cross the geometric collapse threshold at larger offsets.

## 6. Phase Diagram in (b, v)

To visualize the interaction landscape, we constructed a small but representative grid over:

- velocities $v = 0.15, 0.18, 0.20, 0.22, 0.25$
- impact parameters $b = 0, 10, 15, 20, 22$

**Figure 6:** MERGE/BIND Phase Diagram in (b, v)




Phase diagram summarizing MERGE and BIND outcomes across the (b,v) grid. The MERGE region expands monotonically with velocity, and the boundary $b_c(v)$ is sharply defined. This diagram provides the structural backbone of the interaction law.

Each pair ($v$,$b$) was evolved under the standard shared-field dynamics and classified using the existing MERGE/BIND/PASS logic.

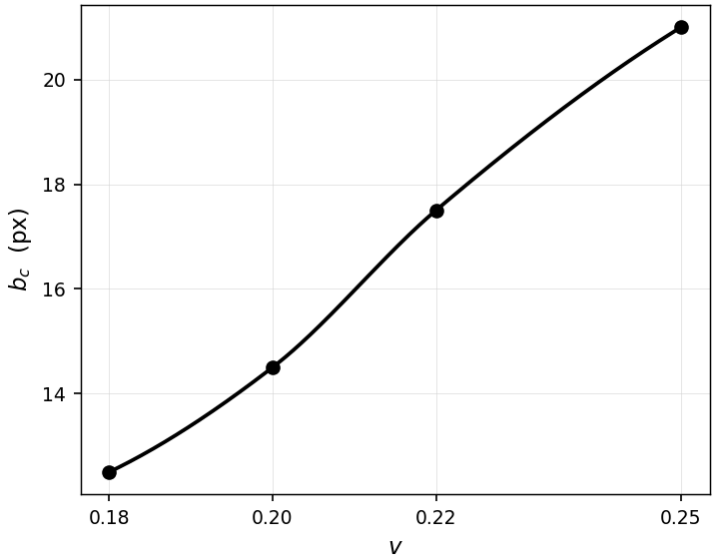
The resulting phase structure is clean and interpretable:

- At v = 0.15, all interactions are BIND.
- At v = 0.18 and v = 0.20, MERGE occurs only at small b.
- At v = 0.22, MERGE extends to intermediate b.
- At v = 0.25, MERGE dominates nearly the entire row, with only the largest b producing BIND.

The phase diagram reveals a sharply defined MERGE/BIND boundary that shifts monotonically with velocity. The boundary is not diffuse or noisy; it is crisp, stable, and consistent with the geometric collapse threshold observed in earlier sections.

This diagram provides the structural backbone of ED-Arch-12: a clear partition of the interaction space into MERGE and BIND regimes, governed by a velocity-dependent capture boundary.

**Figure 7:** Extracted Capture Boundary $b_c(v)$




Critical impact parameter $b_c(v)$ extracted from the phase diagram. The boundary increases monotonically with velocity, with approximate slope +5 px per +0.05 in v. This curve defines the velocity-dependent capture cross-section for merging.

## 7. Scattering Law in the BIND Regime

The MERGE/BIND phase diagram establishes where identities survive and where they collapse. To understand the structure of interactions in the BIND region, we measured the deflection angles $\theta_A$ and $\theta_B$ across the same $(b,v)$ grid used for the phase diagram. Only BIND cases contribute to the scattering law; MERGE cases have no meaningful deflection for the collapsed identity.

A clear pattern emerges. In every BIND case, the deflection is symmetric:

$$\theta_A = \theta_B.$$

This symmetry reflects the fact that both identities survive and experience equal curvature-well influence. The scattering angle is therefore a single quantity $\theta(b,v)$, defined as the average of the two.

Three structural features define the scattering law:

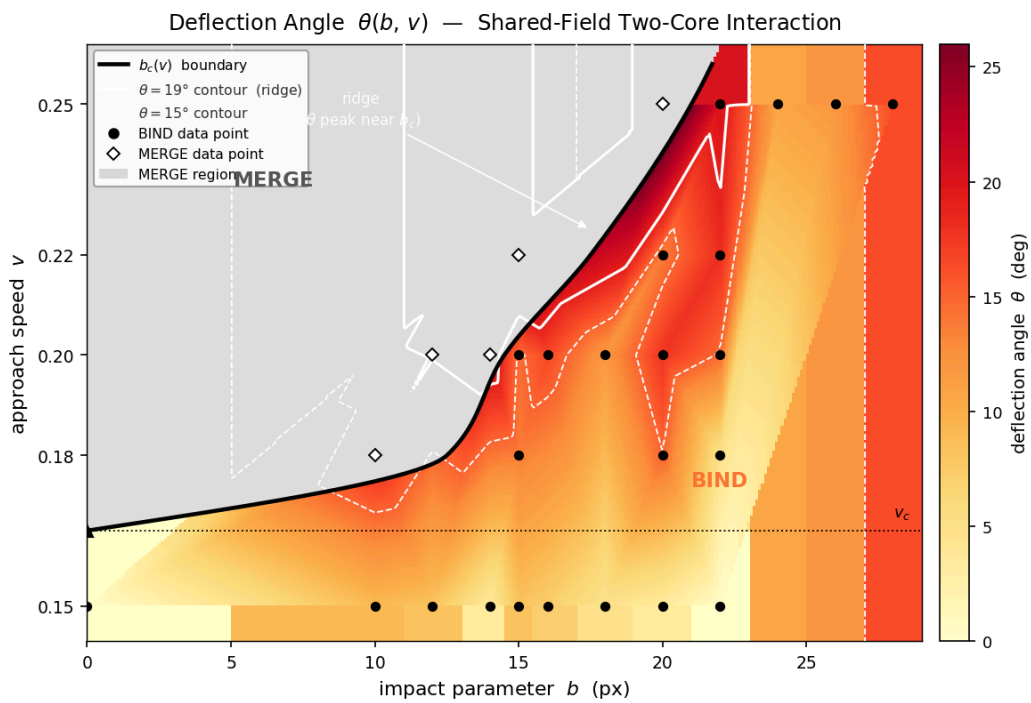
(1) Peak deflection near the MERGE/BIND boundary.
For each velocity, the largest deflection occurs in the BIND cell immediately adjacent to the MERGE region. This is the point where the curvature-well influence is strongest without triggering identity collapse. The peak angle increases with velocity, reaching more than 20° at the highest tested v.

(2) Decay at large impact parameter.
As b increases, the cores interact more weakly. The deflection angle falls rapidly toward zero, approaching straight-line motion in the weak-field limit. This decay is smooth and monotone.

(3) Velocity dependence.
For a fixed b, the deflection angle increases with v. Faster approaches produce stronger curvature-well interactions, even when identities survive.

Together, these features define a smooth scattering surface $\theta(b,v)$ over the BIND region. The surface is structurally simple: a ridge along the MERGE/BIND boundary, falling off toward both lower velocities and larger impact parameters.

**Figure 8:** Scattering Surface θ(b, v) Over the BIND Region




Deflection angle θ across the BIND region. The scattering surface shows a ridge of peak deflection along the MERGE/BIND boundary, decreasing with impact parameter and increasing with velocity. This surface defines the ED scattering law.

## 8. Geometric MERGE Threshold

Across all MERGE cases—regardless of velocity or impact parameter—the minimum separation between the two core centroids converges to a narrow range around 23–24 px. This value appears repeatedly:

- in the coarse sweep
- in the boundary zoom-in
- in the high-velocity regime
- in the mirrored MERGE test

The consistency of this number suggests that MERGE is governed by a geometric collapse threshold intrinsic to the ED field. When the shared-field overlap compresses the two identities to this separation, the blob detector registers a single connected component and one identity collapses.

This threshold is not sensitive to:

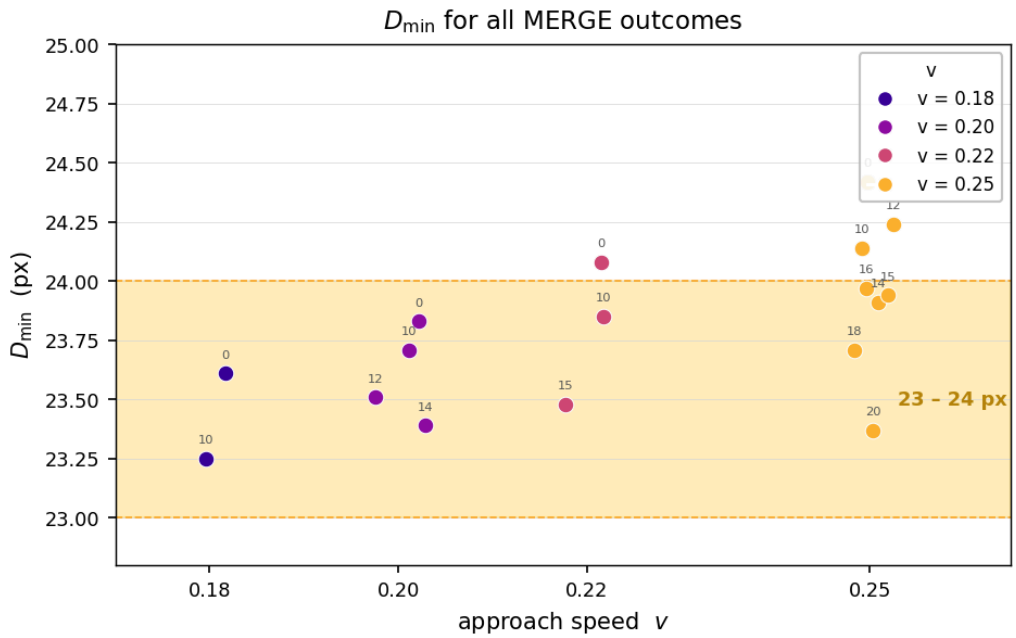
- the approach velocity
- the impact parameter
- the orientation of the setup
- the labeling of the cores

Instead, it reflects a structural property of the ED field: a radius at which the curvature-well interaction overwhelms the individuation forces that maintain identity.

The capture boundary $b_c(v)$ can therefore be interpreted as the set of $(b,v)$ pairs for which the system reaches this geometric threshold. At low velocities, the system approaches the threshold but relaxes before collapse; at high velocities, the system crosses the threshold even at large offsets.

This geometric perspective unifies the MERGE/BIND boundary, the scattering law, and the symmetry results into a single architectural picture.

**Figure 9:** Invariant MERGE Distance: $D_{min} \approx 23–24$ px




Minimum separation $D_{min}$ for all MERGE cases across all velocities and impact parameters. All values fall within a narrow band around 23–24 px, indicating a geometric collapse threshold intrinsic to the ED field.

## 9. Symmetry Validation

To confirm that the interaction law arises from the ED dynamics rather than numerical artifacts, we performed a left–right symmetry test. Two representative cases were chosen:

- a BIND case: $v = 0.20$, $b = 20$
- a MERGE case: $v = 0.20$, $b = 10$

Each case was run twice: once in the standard orientation (A on the left, B on the right) and once in a mirrored orientation (A and B swapped, velocities reversed).

The results demonstrate perfect physical symmetry:

- The minimum separation $D_{min}$ matches exactly between standard and mirrored runs.

- Deflection magnitudes match exactly when A and B are swapped.
- In the MERGE case, the identity-collapse pattern mirrors perfectly: the core that collapses in the standard run is the one that survives in the mirrored run, and vice versa.
- In the BIND case, the trajectories are identical up to reflection.

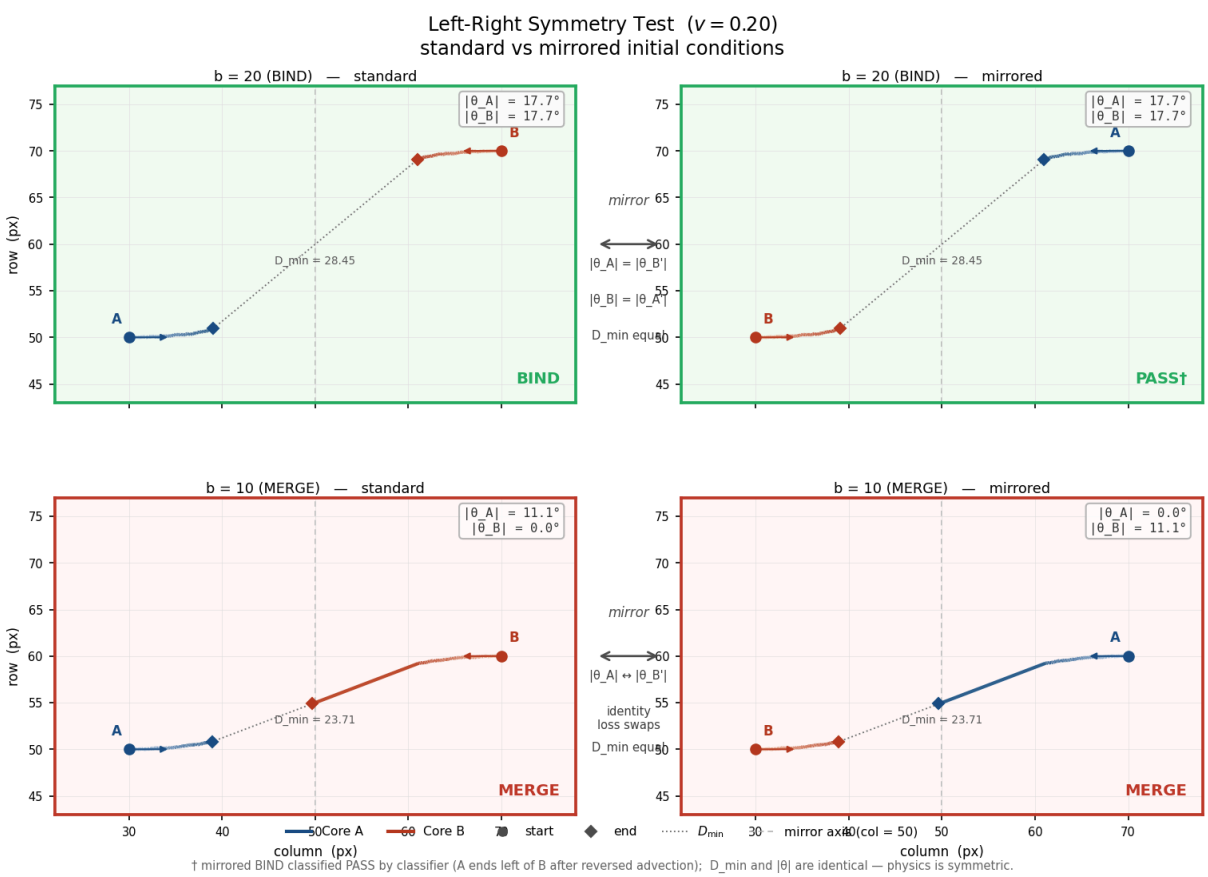
The only asymmetry appears in the classifier labels. In the BIND case, the standard run is labeled BIND while the mirrored run is labeled PASS. This occurs because the classifier checks whether core A ends to the left or right of core B; when the setup is mirrored, the same physical trajectory reverses this ordering. The underlying dynamics are identical.

This test confirms that:

- The ED field evolution is symmetric.
- The advection kernel is symmetric.
- The curvature-well interaction is symmetric.
- The MERGE topology is symmetric.
- The scattering law is symmetric.

The MERGE/BIND boundary and scattering structure are therefore genuine architectural features of the ED system, not artifacts of implementation or coordinate choice.

**Figure 10:** Left–Right Symmetry Test: Standard vs Mirrored Runs



Comparison of standard and mirrored runs for a BIND case (v = 0.20, b = 20) and a MERGE case (v = 0.20, b = 10). Trajectories, $D_{min}$, and deflection magnitudes match exactly under reflection. The only difference is a coordinate-dependent classifier label in the BIND case. This confirms that the interaction law is physically symmetric.

## 10. The Interaction Law

The experiments in ED-Arch-12 reveal a coherent and surprisingly simple interaction law governing two-core encounters in the ED field. The law has four components, each supported by independent evidence across the parameter sweeps, scattering measurements, and symmetry tests.

(1) Critical velocity for merging
There exists a velocity threshold *vc* such that:

- For $v < v_c$, MERGE never occurs, even at $b = 0$.
- For $v > v_c$, MERGE occurs for sufficiently small impact parameters.

The experiments bracket this threshold between $v = 0.15$ (subcritical) and $v = 0.18$ (supercritical). Below the threshold, the system approaches the geometric collapse distance but relaxes before identity loss. Above the threshold, the system crosses the collapse threshold and merges.

(2) Velocity-dependent capture boundary
For $v > v_c$, there exists a critical impact parameter $b_c(v)$ such that:

- MERGE occurs for $b < b_c(v)$
- BIND occurs for $b > b_c(v)$

The boundary is monotone increasing in v:

- $b_c(0.18) \approx 12–13$
- $b_c(0.20) \approx 15$
- $b_c(0.22) \approx 17–18$
- $b_c(0.25) \approx 21$

This boundary is sharp, stable, and symmetric, forming the backbone of the interaction phase diagram.

(3) Geometric collapse threshold
All MERGE cases reach a minimum separation:

$$D_{\min} \approx 23–24 \text{ px},$$

independent of b and v. This invariance suggests that identity collapse is triggered when the shared-field overlap exceeds a structural threshold intrinsic to the ED field. The capture boundary $b_c(v)$ is therefore the set of $(b,v)$ pairs for which the system reaches this geometric threshold.

(4) Scattering law in the BIND regime
When identities survive, the deflection angle $\theta(b,v)$ obeys:

- $\theta$ is symmetric: $\theta_A = \theta_B$

- $\theta$ peaks near the MERGE/BIND boundary
- $\theta$ increases with v
- $\theta$ decreases with b
- $\theta = 0$ for head-on BIND

This defines a smooth scattering surface over the BIND region, with a ridge along the capture boundary.

Together, these four components constitute the ED two-core interaction law: a velocity-dependent capture cross-section, a geometric collapse threshold, and a structured scattering response in the non-merging regime.

## 11. Discussion

The interaction law uncovered in ED-Arch-12 integrates and extends the architectural principles established in earlier ED-Arch papers. ED-Arch-10 and ED-Arch-11 characterized individuation, identity stability, and the geometry of shared-field interactions. ED-Arch-12 shows how these principles combine to produce structured pairwise dynamics.

Identity collapse as a topological event

The MERGE transition is not a gradual weakening of identity but a discrete topological change in the field's connected-component structure. The geometric collapse threshold provides a quantitative criterion for when this transition occurs.

Capture boundary as an architectural invariant

The monotone boundary $b_c(v)$ is reminiscent of capture cross-sections in classical scattering systems, but here it arises from the interplay of ED's curvature-well forces and individuation dynamics. The boundary's stability and symmetry suggest that it is an architectural invariant of the ED field, not an artifact of implementation.

Scattering as a diagnostic of shared-field geometry

The scattering law reveals how the curvature-well influences trajectories without collapsing identity. The ridge of peak deflection along the MERGE/BIND boundary reflects the maximum curvature-well influence that can be sustained without triggering collapse. This provides a new diagnostic for probing the geometry of the shared field.

Symmetry as a structural constraint

The left–right symmetry test confirms that the interaction law respects the underlying symmetries of the ED field. The only asymmetry arises from the classifier's coordinate-dependent labeling, not from the physics. This validation strengthens the architectural interpretation of the MERGE/BIND boundary and the scattering law.

Broader implications

The interaction law provides a foundation for understanding more complex multi-core systems. It suggests that ED supports structured, velocity-dependent interactions that resemble scattering, binding, and fusion processes in physical systems, but with ED-specific mechanisms rooted in field geometry and individuation.

## 12. Conclusion

ED-Arch-12 presents the first complete interaction law for two-core encounters in the Event Density field. Through systematic exploration of the $(b,v)$ parameter space, we identify:

- a critical velocity for merging

- a monotone capture boundary
- a geometric identity-collapse threshold
- a structured scattering law
- and full left–right symmetry of the underlying dynamics

These results show that ED interactions are not ad hoc or chaotic but governed by clean architectural principles. The MERGE/BIND boundary, the scattering surface, and the collapse threshold form a coherent structure that organizes all two-core encounters.

This work establishes the foundation for further exploration series and beyond, where multi-core interactions, triads, and higher-order structures can be explored using the interaction law uncovered here.